
Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Mwesigye, Peter
Date of birth:	9/19/1978
Subject:	050-Meteorology
Result:	55 %
Time used:	00:00:12 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	297	50.	METEOROLOGY	1,00	1,00
2	166	50.	METEOROLOGY	1,00	1,00
3	195	50.	METEOROLOGY	1,00	1,00
4	285	50.	METEOROLOGY	1,00	1,00
5	374	50.	METEOROLOGY	0,00	1,00
6	482	50.	METEOROLOGY	1,00	1,00
7	312	50.	METEOROLOGY	1,00	1,00
8	433	50.	METEOROLOGY	1,00	1,00
9	378	50.	METEOROLOGY	0,00	1,00
10	444	50.	METEOROLOGY	0,00	1,00
11	348	50.	METEOROLOGY	0,00	1,00
12	184	50.	METEOROLOGY	0,00	1,00
13	406	50.	METEOROLOGY	0,00	1,00
14	360	50.	METEOROLOGY	0,00	1,00
15	476	50.	METEOROLOGY	1,00	1,00
16	174	50.	METEOROLOGY	1,00	1,00
17	172	50.	METEOROLOGY	1,00	1,00
18	178	50.	METEOROLOGY	1,00	1,00
19	443	50.	METEOROLOGY	1,00	1,00
20	177	50.	METEOROLOGY	1,00	1,00
21	322	50.	METEOROLOGY	1,00	1,00
22	504	50.	METEOROLOGY	1,00	1,00
23	352	50.	METEOROLOGY	0,00	1,00
24	161	50.	METEOROLOGY	0,00	1,00
25	358	50.	METEOROLOGY	0,00	1,00
26	447	50.	METEOROLOGY	1,00	1,00
27	384	50.	METEOROLOGY	0,00	1,00
28	315	50.	METEOROLOGY	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	491	50. METEOROLOGY		0,00	1,00
30	102	50.1.1.2	Troposphere	1,00	1,00
31	17	50.1.2.5	Development of inversions, types	0,00	1,00
32	113	50.1.6.3	Calculations	0,00	1,00
33	69	50.2.3.1	General circulation around the globe	0,00	1,00
34	81	50.2.4.1	Anabatic and catabatic winds etc	1,00	1,00
35	32	50.2.4.1	Anabatic and catabatic winds etc	0,00	1,00
36	310	50.3	THERMODYNAMICS	1,00	1,00
37	18	50.3.3.1	Adiabatic processes, stability of the atmosphere	1,00	1,00
38	3	50.4.2.2	Radiation fog	1,00	1,00
39	135	50.6.2.2	Warm front, associated clouds & weather	1,00	1,00
40	124	50.8.2.1	Cause and development of tropical showers	0,00	1,00
Total:55%				22,00	40,00

Attachment 2: List of result by topic

Licence

Date: 18.07.2014

Examinee: Mwesigye, Peter

Location: Uganda Civil Aviation Authority

Your Result: 22,00 from 40,00 Points (55,00 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
50. METEOROLOGY	40	22,00	40,00	55,00
50.1. THE ATMOSPHERE	3	1,00	3,00	33,33
50.2. WIND	3	1,00	3,00	33,33
50.3. THERMODYNAMICS	2	2,00	2,00	100,00
50.4. CLOUDS AND FOG	1	1,00	1,00	100,00
50.6. AIR MASSES AND FRONTS	1	1,00	1,00	100,00
50.8. CLIMATOLOGY	1	0,00	1,00	0,00
Total	40	22,00	40,00	55,00

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Date of birth: 1978-09-19

Subject: 050-Meteorology
Date: 2014-07-18



1

How can the stability of the atmosphere be determined?

- ☒ Ambient temperature lapse rate.
- ☐ Surface temperature/dewpoint spread.
- ☐ Atmospheric pressure at various levels.

2

If clouds form as a result of very stable, moist air being forced to ascend a mountain slope, the clouds will be

- ☐ cumulus type with considerable vertical development and turbulence.
- ☐ cirrus type with no vertical development or turbulence.
- ☒ stratus type with little vertical development and little or no turbulence.

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3

Convective currents are most active on warm summer afternoons when winds are

☐ moderate.

☐ strong.

☒ light.

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4

What values are used for Winds Aloft Forecasts?

- ☐ True direction and MPH.
- ☐ Magnetic direction and knots.
- ☒ True direction and knots.

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5

(Refer to Figure 145.) The peak wind at KAMA was reported to be from 320 degrees true at 39 knots,

See Attachments:

1. atp_145.jpg

figure145

(Attachment
No.1)

☒ which occurred at 1743Z.

☐ with .43 of an inch liquid precipitation since the last report.

☒ with gusts to 43 knots.

6

What is indicated by the term 'embedded thunderstorms'?

- ☐ Severe thunderstorms are embedded within a squall line.
- ☒ Thunderstorms are obscured by massive cloud layers and cannot be seen.
- ☐ Thunderstorms are predicted to develop in a stable air mass.

7

How are haze layers cleared or dispersed?

- ☐ By convective mixing in cool night air.
- ☐ By evaporation similar to the clearing of fog.
- ☒ By wind or the movement of air.

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8

Which is an effect of ice, snow, or frost formation on an airplane?

- ☐ Increased pitchdown tendencies.
- ☐ Increased angle of attack for stalls.
- ☒ Increased stall speed.

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9

METAR KSPS 131757Z 09014KT 6SM -RA SCT025 OVC090 24/22 A3005.|SPECI KSPS 131820Z 01025KT 3SM +RA FC OVC015 22/21 A3000.|Which change took place at Wichita Falls (KSPS) between 1757 and 1820 UTC?

☐ Atmospheric pressure increased.

☐ A funnel cloud was observed.

☒ The rain became lighter.

10

Which conditions result in the formation of frost?

- ☐ The temperature of the collecting surface is at or below freezing and small droplets of moisture are falling.
- ☐ Temperature of the collecting surface is below the dewpoint of surrounding air and the dewpoint is colder than freezing.
- ☒ When dew forms and the temperature is below freezing.

11

How will an area of thunderstorm activity, that may grow to severe intensity, be indicated on the Severe Weather Outlook Chart?

☒ SVR within any area.

☐ APCHG within any area.

☐ SLGT within cross-hatched areas.

12

What is the effect of center of gravity on the spin characteristics of an aircraft?

- ☐ If the CG is too far forward, spin entry will be difficult.
- ☒ If the CG is too far aft, spins can become high-speed spirals.
- ☐ A flat spin may develop if the CG is too far aft.

13

(Refer to Figure 151.) The 12-Hour Significant Weather Prognostic Chart indicates that eastern Kentucky and eastern Tennessee can expect probable ceilings

- ☐ less than 1,000 feet and/or visibility less than 3 miles, and moderate turbulence above 10,000 feet MSL.
- ☒ less than 1,000 feet and/or visibility less than 3 miles, and moderate turbulence below 10,000 feet MSL.
- ☐ less than 1,000 feet and/or visibility less than 3 miles.

14

If a SIGMET alert is announced, how can information contained in the SIGMET be obtained?

- ☒ By contacting a weather watch station.
- ☐ ATC will announce the hazard and advise when information will be provided in the FSS broadcast.
- ☐ By contacting the nearest AFSS.

15

Where can wind shear associated with a thunderstorm be found? Choose the most complete answer.

- ☐ In front of the thunderstorm cell and directly under the cell.
- ☒ On all sides of the thunderstorm cell and directly under the cell.
- ☐ In front of the thunderstorm cell (anvil side) and on the right side of the cell.

16

The conditions necessary for the formation of stratiform clouds are a lifting action and

☐ unstable, dry air.

☐ unstable, moist air.

☒ stable, moist air.

17

What type of weather can one expect from moist, unstable air, and very warm surface temperatures?

- ☐ Continuous heavy precipitation.
- ☐ Fog and low stratus clouds.
- ☒ Strong updrafts and cumulonimbus clouds.

18

The formation of either predominantly stratiform or predominantly cumuliform clouds is dependent upon the

☐ temperature of the air being lifted.

☒ stability of the air being lifted.

☐ source of lift.

19

Steady precipitation, in contrast to showers, preceding a front is an indication of

- ☒ stratiform clouds with little or no turbulence.
- ☐ cummuliform clouds with little or no turbulence.
- ☐ stratiform clouds with moderate turbulence.

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20

The presence of standing lenticular altocumulus clouds is a good indication of

☐ lenticular ice formation in calm air.

☐ heavy icing conditions.

☒ very strong turbulence.

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21

Which type precipitation is an indication that supercooled water is present?

☐ Wet snow.

☐ Ice pellets.

☒ Freezing rain.

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22

In what localities is advection fog most likely to occur?

☒ Coastal areas.

☐ Level inland areas.

☐ Mountain slopes.

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23

Most midair collision accidents occur during

☒ **hazy days.**

☐ **cloudy nights.**

☐ **clear days.**

24

Moisture is added to a parcel of air by

☒ **evaporation and condensation.**

☐ **sublimation and condensation.**

☐ **evaporation and sublimation.**

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25

What conditions are indicated on a Weather Depiction Chart?

- ☒ **Actual en route weather conditions between reporting stations.**
- ☐ **Forecast ceilings and visibilities over a large geographic area.**
- ☐ **Actual sky cover, visibility restrictions, and type of precipitation at reporting stations.**

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26

The amount of water vapor which air can hold largely depends on

☐ relative humidity.

☐ stability of air.

☒ air temperature.

27

Constant Pressure Analysis Charts contain contours, isotherms and some contain isotachs. The contours depict

- ☒ ridges, lows, troughs and highs aloft.
- ☐ highs, lows, troughs, and ridges corrected to MSL.
- ☒ highs, lows, troughs, and ridges on the surface.

28

A clear area in a line of thunderstorm echoes on a radar scope indicates

- ☐ the absence of clouds in the area.
- ☐ an area where precipitation drops are not detected.
- ☒ an area of no convective turbulence.

29

What is an important characteristic of wind shear?

- ☐ The Coriolis phenomenon in both high- and low-level air masses is the principal generating force.
- ☐ It is an atmospheric condition that may be associated with a low-level temperature inversion, a jet stream, or a frontal zone.
- ☒ It is an atmospheric condition that is associated exclusively with zones of convergence.

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30

In which layer is most of the atmospheric humidity concentrated ?

☐ Stratosphere.

☐ Tropopause.

☒ Troposphere.

☐ Stratopause.

31

Which of the following statements concerning an isothermal layer in the atmosphere is correct?

- ☐ Air parcels that are forced to rise through an isothermal layer keep the same temperature.
- ☐ An isothermal layer is formed when the air in the layer is well mixed by turbulence.
- ☐ An isothermal layer is absolutely stable.
- ☒ The vertical temperature gradient lies between the dry and saturated adiabatic lapse rates.

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32

An altimeter adjusted to 1013 hPa indicates an altitude of 3600 ft. Should this altimeter be adjusted to the local QNH value of 991 hPa, the altitude indicated would be

☒ 2922 ft

☐ 4194 ft

☐ 4278 ft

☐ 3006 ft

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33

(Please view annex 050-10722A)

Assuming a generalised zonal system of world climatic and wind circulation, zone "t" is an area of

☐ subtropical high pressure systems

☒ travelling low pressure systems

☐ NE trade winds

☐ SE trade winds

See Attachments:

1. 050-10722a.gif

Annex

(Attachment
No.2)

34

A high pressure area (slack pressure gradient) covers part of the Mediterranean Sea and coastal region during the summer. What surface wind direction is likely at an airport at the coast on a sunny afternoon?

☐ Parallel to the coastline.

☒ Sea to land.

☐ Variable.

☐ Land to sea.

35

The sea breeze is a wind

- ☐ that reaches up to the tropopause in daytime
- ☒ occurring only in mid-latitudes and in daytime
- ☐ occurring only in the lower layers of the atmosphere in daytime
- ☐ blowing at night in mid-latitudes

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36

When advection fog has developed, what may tend to dissipate or lift the fog into low stratus clouds?

- ☐ Temperature inversion.
- ☐ Surface radiation.
- ☒ Wind stronger than 15 knots.

37

A parcel of unsaturated air is forced to rise through an isothermal layer. So long as it remains unsaturated, the temperature of the parcel

- ☐ decreases 0.65°C per 100 m
- ☐ remains constant
- ☒ decreases 1°C per 100 m
- ☐ becomes equal to the temperature of the isothermal layer

38

Which of the following is most likely to lead to the dissipation of radiation fog ?

- ☐ Ground cooling caused by radiation during the night
- ☐ A marked decrease in wind velocity close to the ground
- ☒ A marked increase in wind velocity near the ground
- ☐ A build up of a high pressure area resulting in adiabatic warming associated with a sinking air mass

39

The main factor which contributes to the formation of very low clouds ahead of a warm front is the

- ☐ saturation of the warm air by rain falling into it and evaporating
- ☐ warm air moving over a cold surface
- ☒ saturation of the cold air by rain falling into it and evaporating
- ☐ reduction of outgoing radiation due to clouds

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40

The mean height of the tropical zero degree isotherm is

☒ 19000 ft

☐ 16000 ft

☐ 12000 ft

☐ 14000 ft

Attachment No. 1

Questions 5

figure145

atp_145.jpg

AVIATION ROUTINE WEATHER REPORTS (METAR)

TX

METAR KAMA 131755Z 33025G35KT 3/4SM IC OVC003 M02/M01 A2952 RMK PK WND 32039/43 WSHFT 1735 PRESFR P0003.

METAR KAUS 131753Z 19011G17KT 8SM SCT040 BKN250 31/21 A3006 RMK SLPNO.

METAR KBPT 131755Z 17004KT 7SM FEW001 SCT030 BKN250 34/23 A2979 RMK VIS E 2.

METAR KBRO 131755Z 14015KT 6SM HZ SCT034 OVC250 34/30 A2985 RMK PRESRR.

METAR KCDS 131758Z 11013KT 7SM -SHRA OVC180 23/21 A3012 RMK RAB42 VIRGA SW.

METAR KCLL 131749Z 21011KT 7SM SCT003 BKN025 OVC100 34/21 A3008 RMK BKN025 V OVC.

METAR KCOT 131749Z 13010KT 10SM SCT040 SCT200 31/21 A3002 RMK RAE24.

METAR KCRP 131753Z 16016KT 10SM SCT028 BKN250 32/24 A3003.

METAR KDAL 131755Z 16005KT 7SM SCT023 OVC100 30/22 A3007.

METAR KDFW 131800Z 17007KT 10SM SCT035 OVC120 29/20 A3008.

METAR KDHT 131756Z 04014KT 15SM BKN025 22/15 A3026.

METAR KDRT 131756Z 12012KT 10SM FEW006 SCT020 BKN100 OVC250 29/22 A3000 RMK CONS LTG DSTN ESE TS SE MOVG NW VIRGA W.

METAR KERP 131755Z 09007KT 60SM VCBLDU FEW070 SCT170 BKN210 29/13 A3015.

METAR KFTW 131750Z 18007KT 7SM SCT025 OVC100 29/20 A3008.

FTW 131815Z UA /OV DFW/TM 1803/FL095/TP PA30/SK 036 OVC 060/075 OVC/RM TOPS UNKN.

METAR KGGG 131745Z 15008KT 15SM SKC 32/21 A3011.

METAR KGLS 131750Z VRB04KT 6SM VCSH SCT041 BKN093 26/22 A2995.

SPECI KGLS 131802Z 10012G21KT 060V140 2SM +SHRA SCT005 BKN035 OVC050CB 24/23 A2980 RMK RAB57 WSHFT 58 FROPA

METAR KHOU 131752Z 15008KT 7SM SCT030 OVC250 31/27 A3008.

METAR KHRL 131753Z 14015KT 8SM SKC 30/25 A3010.

METAR KIAH 131755Z VRB03KT 1/4SM R33L/1200FT BCFG VV007 27/26 A3005.

METAR KINK 131755Z 04027G36KT 2SM BLSA PO OVC015TCU 24/13 A2985.

METAR KLBB 131750Z 06029G43KT 1SM BLSNDU SQ VV010 03/M01 A2949.

LBB 131808Z UUA /OV LBB /TM 1800 /FL UNKN /TP B737 /TB MDT /RM LLWS -17 KT SFC-010 DURC RWY 36 LBB.

SPECI KLBB 131818Z 35031G40KT 1/2SM FZDZ VV030 M01/M01 A2946 RMK WSHFT 12 FROPA.

LBB 131821Z UUA /OV LBB/TM1817/FL011/TP B727/SK UNKN OVC/TA -06 /TB MOD/IC MDT CLR.

METAR KLFK 131756Z 24007KT 7SM BKN100 33/19 A3008.

METAR KMAF 131756Z 02020KT 12SM BKN025 OVC250 27/18 A3009 RMK RAE44.

METAR KMFE 131756Z 13015KT 7SM BKN125 33/19 A2998.

METAR KMRF 131752Z 09012G20KT 60SM SKC 28/14 A3000.

MRF 131801Z UUA/OV MRF/TM1758/FL450/TP B767/TB MDT CAT.

Figure 145. Aviation Routine Weather Reports (METAR) © ASA

Attachment No. 2

Questions 33

Annex

050-10722a.gif

050 - 10722 A

